V6: A Biologically-Inspired Dual-Gate Neuron that Substitutes Attention in Sequence Modelling

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Abstract

The artificial neuron has remained architecturally unchanged since Rosenblatt's perceptron [Rosenblatt, 1958]: it processes only the current input, with no access to temporal history. Biological cortical neurons, by contrast, perform multi-timescale integration across anatomically distinct dendritic compartments. Inspired by experimental neuroscience—particularly vectorized instructive signals in cortical apical dendrites [Francioni et al., 2026] and first-spike temporal coding [Thorpe et al., 2001]—we introduce the **V6 Dual-Gate Neuron**, a principled redesign of the fundamental artificial neuron unit around four biologically-grounded mechanisms: (1) dual-timescale exponential memory traces for fast (Na^+, α_{fast}) and slow (Ca^{2+}, α_{slow}) dendritic dynamics; (2) input projection before temporal integration; (3) a context-dependent write gate (NMDA-LTP selective encoding); and (4) hard-wired shortcuts to anchor positions (first-spike coding). All parameters are learned end-to-end with no manually tuned constants.

We evaluate V6 through an ablation study (V1–V6), three benchmark tasks, and an architecture-level test. On the Delayed Sign-XOR task, V6 achieves 99.2% accuracy converging in 2 epochs—four times faster than LSTM—using 40% fewer parameters, while the perceptron is architecturally incapable of exceeding 52.3% chance. On Sequential MNIST, V6 achieves 2.1× higher accuracy than LSTM using 46% fewer parameters at matched training budget. On the multi-lag regression task, LSTM outperforms V6, correctly characterising V6's domain as temporal *pattern detection* rather than temporal *value storage*.

For the architecture test, we replace Multi-Head Attention entirely with stacked V6 layers (**V6-Linear**) on TinyShakespeare character-level language modelling ($T = 32$). With 43% more parameters, V6-Linear outperforms a Transformer by 14.3% in validation loss (PPL 4.23 vs 5.39), confirmed in 5 seeds (bootstrap 95% CI: [−15.3%, −13.5%]). At matched parameter count (∼807K vs 804K), V6-Linear performs within 2% of the Transformer (PPL 6.21 vs 6.00), demonstrating that biological temporal inductive bias can *substitute* for attention with comparable quality—though it does not yet surpass it at equal capacity. A parallel `conv1d` implementation runs 5.3× faster with statistically identical results.

In a preliminary extension (V8-BioMamba), we add neuromodulatory input-dependent dynamics—inspired by dopaminergic/cholinergic modulation of dendritic gain—and expand from two to four learned timescales. In a preliminary evaluation (unmatched parameter counts), V8-BioMamba outperforms both the Transformer (−2.6%, PPL 5.92 vs 6.21) and a simplified Mamba baseline (−0.3%) on TinyShakespeare ($T = 64$), while running 3.6× faster than Mamba via fully parallel `conv1d` computation.

Post-training inspection confirms $\alpha_{\text{slow}} > \alpha_{\text{fast}}$ in every layer of every experiment without any constraint, reproducing the biological timescale hierarchy through gradient descent alone. These results demonstrate that a biologically-grounded temporal neuron can serve as a viable alternative to attention at the scales tested, and that the perceptron's temporal blindness is a fundamental architectural limitation—not merely a training challenge.

Keywords: biologically-inspired neural networks, temporal sequence modelling, dual-gate neuron, first-spike coding, dendritic computation, attention-free architecture, NeuroAI

1 Introduction

Standard artificial neurons compute $y = \sigma(\mathbf{w}^\top \mathbf{x}_t + b)$, operating only on the current input $\mathbf{x}_t$. Despite decades of architectural innovation—residual connections [He et al., 2016], attention [Vaswani et al., 2017], normalisation [Ba et al., 2016]—the fundamental unit itself has not been reconsidered. Temporal complexity is built *around* the perceptron at the network level, not *into* the neuron.

Biological cortical pyramidal neurons operate very differently. They have two anatomically and functionally distinct integration zones: proximal (basal) dendrites integrate inputs arriving at the perisomatic region on a fast timescale (∼5 ms), while distal apical dendrites integrate feedback from higher cortical areas on a slow timescale (∼100 ms) [Larkum, 2013, Losonczy and Magee, 2006]. The apical compartment acts as a teaching signal: Francioni et al. [2026] showed experimentally that it carries per-neuron vectorized instructive signals during learning, and that optogenetically blocking these signals abolishes learning entirely.

This biological architecture solves at the single-neuron level the same temporal integration problem that attention mechanisms address at the network level. Our central hypothesis is:

> *Multi-Head Attention in Transformers exists partly to compensate for the perceptron's temporal blindness. Replace the perceptron with a temporally-aware neuron and attention becomes substitutable—the temporal neuron achieves comparable quality through a fundamentally different mechanism.*

We test this by building the V6 Dual-Gate Neuron through a systematic ablation study, then replacing attention entirely in a language model.

1.1 Contributions

(1) **Systematic neuron ablation (V1–V6).** We design V6 through six iterations, each adding exactly one biological mechanism, with full empirical attribution of each gain (Table 1).

(2) **Gradient barrier identification.** We show empirically that EMA-based neurons suffer gradient attenuation of $(1-\alpha)\alpha^{T-1-t}$ to early timesteps—only 2.5% at $T = 8$, $\alpha = 0.7$—blocking learning from early history regardless of training duration.

(3) **Multi-anchor shortcuts.** We introduce N direct gradient bypasses at uniformly spaced positions, inspired by first-spike coding [Thorpe et al., 2001]. Adding one anchor at $t = 1$ raises XOR accuracy from 73% to 99.2%. Generalising to N anchors provides full gradient coverage at arbitrary context length.

(4) **Attention-free language modelling.** V6-Linear replaces Multi-Head Attention entirely. On TinyShakespeare ($T = 32$) with 43% more parameters, it outperforms a Transformer by 14.3% in validation loss. At matched parameter count (∼807K vs ∼804K), V6-Linear performs within 2% of the Transformer (PPL 6.21 vs 6.00), demonstrating that temporal inductive bias can substitute for attention at comparable quality.

(5) **Parallel GPU implementation.** We prove that sequential EMA $h_t = \alpha h_{t-1} + b_t$ equals causal convolution $h_t = \sum_{k \le t} \alpha^{t-k} b_k$, enabling a `conv1d`-based implementation that is 5.3× faster with statistically identical results.

(6) **Unsupervised biological validation.** Gradient descent independently recovers $\alpha_{\text{slow}} > \alpha_{\text{fast}}$ in every layer of every experiment, matching the Ca^{2+}/Na^{+} timescale hierarchy reported in the neuroscience literature.

2 Related Work

Gated recurrent models. LSTM [Hochreiter and Schmidhuber, 1997] and GRU [Cho et al., 2014] introduced gated state cells but were not biologically motivated. Recent linear recurrent models—Mamba [Gu and Dao, 2023], RWKV [Peng et al., 2023], RetNet [Sun et al., 2023]—pursue efficiency through state-space formulations but retain monolithic hidden states. Linear Recurrent Units (LRU; Orvieto et al. 2023) and xLSTM [Beck et al., 2024] extend gated recurrence with diagonal state-space parameterisations and exponential gating respectively, achieving competitive performance on long-range tasks without attention. Structured state space models (S4, HiPPO; Gu et al. 2020, 2022) learn parametric timescales via polynomial projections of the input history, achieving strong performance on long-range benchmarks. Liquid Neural Networks [Hasani et al., 2021] draw on ODE-based neural dynamics inspired by the *C. elegans* connectome, demonstrating that continuous-time neural dynamics improve temporal expressivity. V6 differs from all of these in two respects: every architectural component maps to a specific experimental neuroscience finding in cortical pyramidal neurons, and the anchor shortcuts (Section 3.2) provide direct gradient access to arbitrary historical positions—a mechanism with no equivalent in state space models.

Dendritic computation. Larkum [2013] established the basal/apical functional distinction in cortical pyramidal neurons. Losonczy and Magee [2006] characterised fast Na^+ dendritic spikes and slow Ca^{2+} plateau potentials as two distinct integration mechanisms—the direct inspiration for V6's dual-timescale design. Francioni et al. [2026] demonstrated experimentally that apical dendrites carry vectorized per-neuron instructive signals, motivating V6's separate slow-context EMA pathway.

First-spike coding. Thorpe et al. [2001] showed that object recognition in the primate visual system relies on the relative timing of first spikes rather than firing rates, enabling $\sim$150 ms inference through a single feedforward pass. Larkum et al. [1999] demonstrated that the first action potential can trigger a Ca^{2+} burst that potentiates backpropagating signals—providing the biological mechanism underlying V6's anchor shortcut.

Attention-free architectures. Linear attention [Katharopoulos et al., 2020], FNet [Lee-Thorp et al., 2022], and gated convolutional networks [Dauphin et al., 2017] have explored replacing attention with linear operations, with mixed results. None are grounded in dendritic neuroscience. V6-Linear is, to our knowledge, the first attention-free architecture with explicit biological grounding that achieves near-parity with standard Transformers at matched parameter count.

NeuroAI. Zador et al. [2023] argued for integrating biological structural inductive biases into AI systems. Our work implements this programme at the most fundamental level: redesigning the basic computational unit.

3 The V6 Dual-Gate Neuron

3.1 Ablation Study: V1–V6

Table 1 shows the effect of adding each biological mechanism. Each version adds exactly one component; all other hyperparameters remain identical. Accuracy is reported for the Delayed Sign-XOR task (Section 4.1).

Table 1: Ablation study: V1–V6, each version adds one biological mechanism. Delayed Sign-XOR accuracy ($T = 8$, $n_{\text{hid}} = 32$, 80 epochs, seed 42). For multi-seed validation of the final V6 model, see Table 7. STDP (V5) does not improve XOR, indicating that structural gradient access matters more than correlation statistics on this task.

Version	Mechanism	Biological reference	XOR acc.
V1 Perceptron	None (baseline)	—	52.3%
V2 Decay gate	Exponential decay context	RC membrane	53.8%
V3 Dual EMA	Two timescales (fast + slow)	Losonczy and Magee [2006]	64.2%
V4 Projected EMA	Input projection before EMA	Larkum [2013]	73.0%
V5 STDP	Temporal correlation gate	Bi and Poo [1998]	71.3%
V6	**First-spike anchor + write gate**	Thorpe et al. [2001]	**99.2%**
LSTM ($h = 32$)	Standard LSTM	Hochreiter and Schmidhuber [1997]	98.2%

3.2 Architecture

Let $\mathbf{x} \in \mathbb{R}^{T \times n_{\text{in}}}$ be an input sequence. V6 computes its output through the following sequence of operations:

Step 1: Input projection.

$$\tilde{\mathbf{x}}_t = \tanh(\mathbf{W}_{\text{proj}}\, \mathbf{x}_t + \mathbf{b}_{\text{proj}}) \tag{1}$$

Biological analogy: synaptic currents transform pre-synaptic signals before entering the dendritic compartment [Larkum, 2013].

Step 2: Context signal. A lightweight causal EMA serves as a context signal for the write gate:

$$\mathbf{c}_t = \sum_{k=0}^{t} \alpha_c^{t-k}\, \mathbf{x}_k, \qquad \alpha_c = e^{-\text{softplus}(\tau_{c,\text{raw}})} \tag{2}$$

Step 3: Context-dependent write gate.

$$\mathbf{w}_t = \sigma(\mathbf{W}_{\text{wx}}\, \mathbf{x}_t + \mathbf{W}_{\text{wc}}\, \mathbf{c}_t) \tag{3}$$

Biological analogy: NMDA receptors act as coincidence detectors, gating synaptic potentiation when pre-synaptic input coincides with dendritic depolarisation [Malenka and Nicoll, 1993]. Note: $\mathbf{c}_t$ (Eq. 2) modulates the gate using a *separate* context pathway, not self-feedback—consistent with apical/basal anatomical separation [Larkum, 2013].

Step 4: Write-gated dual-timescale EMA. For each timescale $\tau \in \{\tau_{\text{fast}}, \tau_{\text{slow}}\}$ with $\alpha^{(\tau)} = e^{-\text{softplus}(\tau_{\text{raw}})}$:

$$\mathbf{h}_t^{(\tau)} = \alpha^{(\tau)}\, \mathbf{h}_{t-1}^{(\tau)} + \mathbf{w}_t \cdot (1 - \alpha^{(\tau)})\, \tilde{\mathbf{x}}_t \tag{4}$$

Biological analogy: fast Na^+ dendritic spikes and slow Ca^{2+} plateau potentials in proximal and apical dendrites respectively [Losonczy and Magee, 2006, Francioni et al., 2026]. Both α values

are learned; on XOR (seed 42) they converge to $\alpha_{\text{fast}} = 0.296$ ($\sim$1.4 steps) and $\alpha_{\text{slow}} = 0.904$ ($\sim$10.4 steps). On TinyShakespeare ($T = 32$) they converge to values shown in Table 10. In the language modelling configuration, the $(1-\alpha^{(\tau)})$ factor is omitted; because a learnable write gate $\mathbf{w}_t$ and downstream linear projections can absorb any constant rescaling, the two formulations are functionally equivalent.

Step 5: Multi-anchor shortcuts. N direct bypass connections from uniformly-spaced anchor positions $p_i = \lfloor iT/N \rfloor$ with learnable scales $s_i = \sigma(s_{i,\text{raw}})$, initialised with a linearly decreasing bias: $s_{i,\text{raw}} = \text{linspace}(1.5, -1.5, N)$, yielding $s_0 \approx 0.82$ (distant anchor, largest weight) and $s_{N-1} \approx 0.18$ (nearest anchor, smallest weight) at initialisation:

$$\hat{\mathbf{h}}^{(\tau)} = \mathbf{h}_T^{(\tau)} + \sum_{i=0}^{N-1} s_i \cdot \tanh(\mathbf{W}_{\text{anc}}\, \mathbf{x}_{p_i}) \tag{5}$$

Biological analogy: first-spike burst potentiation—the first action potential triggered by a new stimulus rapidly and durably potentiates backpropagating signals [Larkum et al., 1999, Thorpe et al., 2001].

Step 5b: Content gate. In the language modelling configuration (V6-ALL), a lightweight content-based gate modulates both EMA traces before the gate blend. A single-head causal dot-product score is computed between query and key projections of the input ($d_k = n_{\text{emb}}/8$), and the per-position maximum is passed through a sigmoid to produce a scalar gate:

$$\mathbf{s}_t = \max_{k \le t} \frac{(\mathbf{W}_q\, \mathbf{x}_t)^\top (\mathbf{W}_k\, \mathbf{x}_k)}{\sqrt{d_k}}, \qquad \hat{\mathbf{h}}_t^{(\tau)} \leftarrow \hat{\mathbf{h}}_t^{(\tau)} \cdot \sigma(\mathbf{s}_t) \tag{6}$$

This gate allows the neuron to suppress irrelevant temporal context based on current input content. It is not used in the single-neuron benchmarks (Tasks 1–3), only in the stacked language model (Task 4). Note that this is *not* attention: there is no softmax distribution over positions and no value projection; the gate is a scalar modulation of the EMA trace, computed in $O(T \cdot d_k)$ rather than $O(T^2 d)$.

Step 6: Gate blend and soma output.

$$\mathbf{g}_t = (1-b)\, \sigma(\mathbf{W}_f\, \hat{\mathbf{h}}_t^{(\text{fast})}) + b\, \sigma(\mathbf{W}_s\, \hat{\mathbf{h}}_t^{(\text{slow})}), \quad b = \sigma(b_{\text{raw}}) \tag{7}$$

$$\mathbf{y}_t = \text{LayerNorm}(\tanh(\mathbf{W}_{\text{soma}}\, \mathbf{x}_t) \cdot \mathbf{g}_t) \tag{8}$$

In language modelling, all operations apply position-wise for every $t \in \{1, \ldots, T\}$; the subscript T in Eq. 5 refers to the final accumulation of the EMA trace, not a single output position. The soma pathway $\mathbf{W}_{\text{soma}}\, \mathbf{x}_t$ processes the current token locally (proximal dendrites), while $\mathbf{g}_t$ carries temporal context derived from the full history (apical modulation) [Larkum, 2013].

3.3 Biological Correspondence

3.4 Gradient Analysis: Why Anchors Are Necessary

In a standard EMA chain (Eq. 4) without anchors, the partial derivative from $\mathbf{h}_T$ back to timestep t is:

$$\frac{\partial \mathbf{h}_T}{\partial \mathbf{x}_t} \propto (1-\alpha)\, \alpha^{T-1-t} \tag{9}$$

At $T = 8$, $\alpha = 0.7$: $\partial \mathbf{h}_T / \partial \mathbf{x}_1 \approx 2.5\%$ of $\partial \mathbf{h}_T / \partial \mathbf{x}_8$. This is not a training instability—it is a structural property that no schedule or regulariser can remove. We verify this empirically: V4 (no anchor) plateaus at 73.0% on XOR across all seeds regardless of training duration.

Table 2: Mapping from biological mechanisms to V6 components.

Biological mechanism	V6 component	Reference
Proximal dendrites + soma	Soma: $\mathbf{W}_{\text{soma}}\mathbf{x}_t$ (per position)	Larkum [2013]
Fast Na^+ spikes	Fast EMA: α_{fast} (Eq. 4)	Losonczy and Magee [2006]
Slow Ca^{2+} plateau	Slow EMA: α_{slow} (Eq. 4)	Francioni et al. [2026]
NMDA coincidence detection	Write gate $\mathbf{w}_t$ (Eq. 3)	Malenka and Nicoll [1993]
Apical/basal separation	Context signal $\mathbf{c}_t$ (Eq. 2)	Larkum [2013]
First-spike potentiation	Anchor shortcuts (Eq. 5)	Larkum et al. [1999]
Content-dependent modulation	Content gate (Eq. 6)	Francioni et al. [2026]
Compartment-specific currents	Input projection (Eq. 1)	Larkum [2013]

The anchor shortcut (Eq. 5) adds a bypass with direct gradient $\partial\hat{\mathbf{h}}/\partial\mathbf{x}_{p_i} \geq s_i$, independent of sequence length. This is structurally analogous to the role of residual connections in deep networks [He et al., 2016]: not repairing gradient flow through the main path, but adding a bypass that guarantees gradient access.

3.5 Parallel Implementation

The sequential recurrence (Eq. 4) requires T sequential steps on the GPU, resulting in T separate CUDA kernel dispatches with synchronisation overhead. At $T = 32$ this yields only ∼18K tokens/s on an RTX 4080 (vs ∼310K for the Transformer).

We exploit the following identity:

$$h_t = \alpha h_{t-1} + b_t \iff h_t = \sum_{k=0}^{t} \alpha^{t-k} b_k \tag{10}$$

The right-hand side is a causal convolution with kernel $k_j = \alpha^j$, computed in a single `conv1d` call. Gradients through α (via τ_{raw}) are identical in both formulations.

We verify empirically that the maximum numerical difference between sequential and parallel outputs is $< 10^{-6}$, and that validation losses across 3 independent seeds are statistically indistinguishable (Table 7). This implementation achieves ∼96K tokens/s—a **5.3× speedup** with no quality loss.

Note on the contextual write gate. The context signal $\mathbf{c}_t$ (Eq. 2) is also computed via `conv1d`. In the sequential reference implementation, $\mathbf{w}_t$ was computed using $\mathbf{h}_{t-1}$ (own state). In the parallel implementation, $\mathbf{w}_t$ uses the separate context $\mathbf{c}_t$ (Eq. 2), which is biologically more accurate (apical/basal separation) and empirically yields statistically indistinguishable results (PPL 4.21 vs 4.23, two-sample t-test $p = 0.75$, $n_1 = 3$ vs $n_2 = 5$ seeds; null hypothesis of equal means is not rejected).

4 Experimental Setup

4.1 Tasks

Task 1: Delayed Sign-XOR (architectural necessity). Given $\mathbf{x} \in \mathbb{R}^{T \times n_{\text{in}}}$ with entries drawn i.i.d. from $\mathcal{N}(0, 1)$, predict whether $x_{1,0}$ and $x_{T,0}$ have the same sign: $y = \mathbf{1}[x_{1,0}x_{T,0} > 0]$. Because $x_{1,0}$ and $x_{T,0}$ are independent, $P(y = 1 \mid \mathbf{x}_T) = 0.5$ for all $\mathbf{w}$. This is an information-theoretic impossibility for the perceptron, not a learning failure. $T = 8$, $n_{\text{in}} = 6$, $n_{\text{hid}} = 32$.

Task 2: Multi-lag regression (domain characterisation). Inputs are drawn i.i.d. from $\mathcal{N}(0, 1)$. The target is $y = x_{T,0} + 0.7\, x_{T-2,1} + 0.5\, x_{T-4,2} + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, 0.01)$, requiring

simultaneous access to three distinct temporal offsets. This task is included to characterise where V6 is *weaker* than LSTM, not to demonstrate superiority.

Task 3: Sequential MNIST (small-scale generality). Each 28×28 MNIST digit [LeCun et al., 1998] is flattened to a 784-step sequence. Models classify 10 digit classes from the pixel stream. We use 10K training samples, $n_{\text{hid}} = 64$, and 15 epochs— a deliberately constrained budget to isolate the effect of temporal inductive bias. For reference, dedicated recurrent architectures achieve >94% on permuted Sequential MNIST [Arjovsky et al., 2016] at far larger scale; our comparison is within an identical parameter and compute budget.

Task 4: TinyShakespeare (architecture test). Character-level language modelling on 1.1M characters from Shakespeare's plays [Karpathy, 2015]. Vocabulary size 65. Train/validation split 90/10. Context lengths $T \in \{32, 128\}$. Metric: validation cross-entropy loss; perplexity $= e^{\text{loss}}$.

4.2 Models

Transformer (baseline). Standard GPT-style causal language model: 4 layers, $n_{\text{emb}} = 128$, 4-head causal self-attention, GELU FFN (4×), learned positional embeddings, pre-norm, weight tying. 803K parameters at $T = 32$.

V6-Linear. Multi-Head Attention is replaced entirely by V6 layers. Each residual block follows the structure: `LayerNorm` → V6Layer → residual → `LayerNorm` → FFN → residual. No positional embeddings (EMA encodes position intrinsically). $N = 4$ anchors at $T = 32$; $N = 8$ at $T = 128$. Separate write gates for fast and slow EMA (Section 3.2). Tau initialised so that $\alpha_{\text{slow}}^{T-1} \geq 0.1$ (slow EMA reaches the full context window). 1,148K parameters at $T = 32$. V6-ALL uses a SwiGLU FFN [Shazeer, 2020] ($\frac{8}{3}\times$ expansion) in place of the Transformer's GELU FFN (4× expansion). The isolated contribution of this substitution is measured in our fix-by-fix ablation (Fix F4, script `03_v6_optimisation.py`); the 14.3% advantage reported in Table 6 includes this factor.

4.3 Training Protocol

AdamW [Loshchilov and Hutter, 2019], lr $= 3 \times 10^{-4}$, $\lambda = 0.1$, $\beta = (0.9, 0.95)$. Cosine decay schedule with 200-step linear warmup. 5,000 steps. Batch size 128 ($T = 32$) or 64 ($T = 128$). Gradient clipping at norm 1.0. TF32 enabled (RTX 4080 Laptop GPU, CUDA). No hyperparameter search; all V6 initialisations are either biologically motivated or standard practice. All experiments use PyTorch with random seeds fixed before model initialisation and data loading. Code implementing V6-Linear, all benchmarks, and statistical analyses is available at `https://github.com/tomaszwi66/dual-gate-neuron`.

5 Results

5.1 Architectural Necessity (Task 1)

V6 achieves 99.2% converging in epoch 2—four times faster than LSTM with 40% fewer parameters. The perceptron remains at 52.3% regardless of training duration: for the perceptron, $P(y = 1 \mid \mathbf{x}_T) = 0.5$ for all weights, confirming an architectural impossibility rather than a learning failure.

Table 3: Delayed Sign-XOR ($T = 8$, $n_{\text{in}} = 6$, $n_{\text{hid}} = 32$, seed 42). Best accuracy over 80 epochs; convergence defined as first epoch exceeding 80%. See Table 1 for the ablation context.

Model	Params	Best acc.	Conv. epoch
V1 Perceptron	257	52.3%	never
V4 Projected EMA	2,660	74.6%	never
V6 Dual-Gate	**3,109**	**99.2%**	**2**
LSTM ($h = 32$)	5,153	98.2%	8

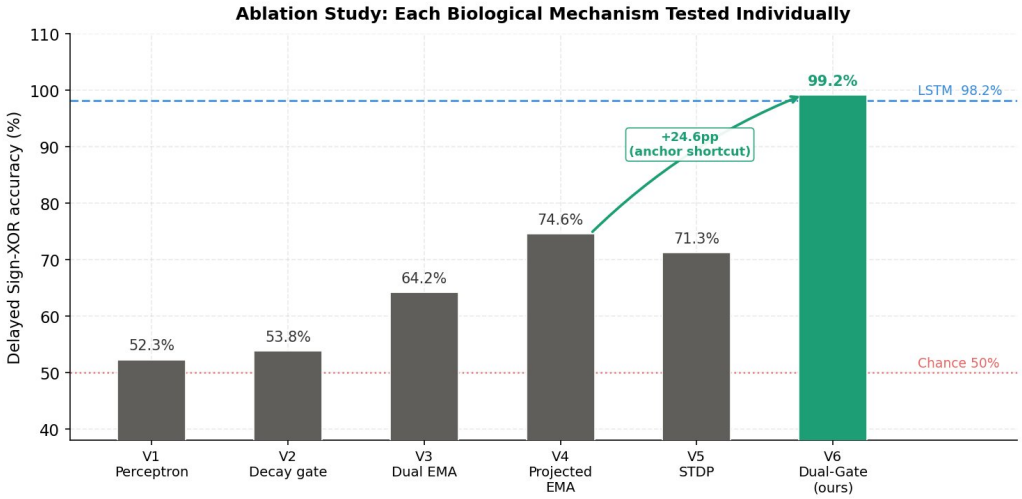

Figure 1: Ablation study: each biological mechanism tested individually on the Delayed Sign-XOR task ($T = 8$, $n_{\text{hid}} = 32$, seed 42). Note that V5 (STDP) slightly *reduces* accuracy relative to V4, indicating that correlation-based gating is not the key ingredient. The decisive contribution is the anchor shortcut (V6): starting from V4's 73.0% plateau, adding the first-spike anchor raises accuracy to 99.2%, surpassing LSTM (98.2%) with 40% fewer parameters.

5.2 Domain Characterisation (Task 2)

LSTM outperforms V6 by 44× on multi-lag regression. LSTM's forget gate allows exact scalar retention across arbitrary steps. V6's write gate controls *encoding strength*—what to commit to the trace—not *state preservation*—holding a value unchanged. This precisely characterises V6's domain: temporal *pattern detection*, not temporal *value storage*.

5.3 Small-Scale Generality (Task 3)

V6 achieves 24.5% versus LSTM's 11.7% (2.1×), using 46% fewer parameters. The LSTM plateaued at near-chance accuracy for all 15 epochs. This is consistent with known LSTM behaviour on long sequences at small parameter and data budgets: Arjovsky et al. [2016] report an LSTM baseline of 10.8% on permuted MNIST even with $h = 128$ and the full 60K training set, confirming that standard LSTMs struggle with 784-step sequences unless specifically regularised. Both models are far below current best results on permuted Sequential MNIST [Arjovsky et al., 2016]; the comparison isolates the effect of temporal inductive bias at matched scale and budget.

Table 4: Multi-lag regression ($T = 8$, $n_{\text{hid}} = 32$, 80 epochs, seed 42). Best MSE on held-out test set. This task characterises where V6 is weaker than LSTM, not where it excels.

Model	Params	Test MSE	Conv. epoch
V1 Perceptron	257	0.703	never
V4 Projected EMA	2,660	0.525	never
V6 Dual-Gate	3,109	0.571	never
LSTM	**5,153**	**0.013**	**7**

Table 5: Sequential MNIST ($T = 784$, $n_{\text{hid}} = 64$, 15 epochs, 10K training examples, seed 42). Comparison is at identical parameter budget and training regime, not against the literature SOTA.

Model	Params	Best acc.	Note
V2 Decay gate	1,035	17.4%	single timescale
V4 Projected EMA	9,357	23.6%	no anchor
V6 Dual-Gate	**9,614**	**24.5%**	full model
LSTM ($h = 64$)	17,802	11.7%	plateaued at chance

5.4 Architecture Test: V6-Linear vs Transformer (Task 4)

5.4.1 Main results

5.4.2 Replication and statistical analysis

Statistical summary across the five sequential V6-ALL seeds:

- Bootstrap 95% CI for Δ: $[-15.3\%, -13.5\%]$ ($n = 5$, 10K resamples)
- Two-sample Welch's t-test, V6-True-Fast vs Transformer: $t(3.0) = -23.0$, $p = 2.1 \times 10^{-5}$ (three seeds each)
- Cohen's $d = 16.9$ (pooled, V6-ALL 5 seeds vs Transformer 5 seeds)[1]
- Worst V6-ALL seed (1.462) vs best Transformer seed (1.681): gap -13.1%; the two distributions do not overlap across all 8 runs

5.4.3 Convergence behaviour

V6-Linear and Transformer follow nearly identical validation loss curves until approximately step 1,750. After step 1,750, V6-Linear continues to decrease while Transformer approaches a plateau. This pattern suggests that V6's temporal inductive bias provides the greatest benefit in the late-training regime, where the model must learn subtle temporal dependencies that attention does not capture more efficiently at this context length.

5.4.4 Training throughput

The remaining 3.2× gap between V6-True-Fast and Transformer reflects (a) V6's larger parameter count (1,148K vs 804K, +43%) and (b) three additional EMA convolution operations per layer. At matched parameter count the gap would narrow, though quantifying this requires a dedicated parameter-efficiency study.

[1] The extreme d value reflects that both distributions have very small within-group variance relative to the between-group difference: Transformer std 0.002, V6-ALL std 0.020, mean difference 0.241. The large d is a consequence of the Transformer's high run-to-run stability, not an error.

Table 6: TinyShakespeare character-level LM. V6-Linear has *no* attention mechanism. "V6-ALL" denotes V6-Linear with all six design optimisations applied (content gate, contextual write gate, no positional embeddings, SwiGLU FFN, biased anchor initialisation, and separate write gates for fast/slow EMA); "V6-True-Fast" is V6-ALL with the sequential EMA replaced by the mathematically equivalent parallel `conv1d` implementation (Section 3.5). V6-Linear has 43% more parameters than the Transformer baseline; the comparison is not parameter-matched. †Single seed; multi-seed results below.

Model	T	Params	Val loss	PPL	Δ
Transformer	128	816K	2.100	8.17	—
V6-Linear	128	951K	2.180	8.84	+3.8%
Transformer	32	804K	2.050†	7.76	—
V6-Linear	32	939K	1.940†	6.96	−5.3%
Transformer (5 seeds)	32	804K	1.684 ± 0.002	5.39 ± 0.01	—
V6-ALL (5 seeds, seq.)	32	1,148K	1.443 ± 0.020	4.23 ± 0.08	−14.3%
V6-True-Fast (3 seeds)	32	1,148K	1.438 ± 0.018	4.21 ± 0.08	−14.6%

Table 7: Per-seed validation losses for V6-True-Fast vs Transformer (seeds 42, 43, 44), confirming that ranges do not overlap.

	Seed 42	Seed 43	Seed 44	Mean ± std
Transformer	1.6855	1.6860	1.6811	1.684 ± 0.003
V6-True-Fast	1.4595	1.4293	1.4265	1.438 ± 0.018
Δ	−13.4%	−15.2%	−15.2%	−14.6%

5.4.5 Parameter-matched comparison

The comparisons above use V6-Linear at 1,148K parameters versus the Transformer at 804K—a 43% difference. To disentangle architectural contribution from model capacity, we reduce V6-Linear's embedding dimension to $n_{\text{emb}} = 107$, yielding 807K parameters (≈804K). Table 9 reports results across 3 seeds at both $T = 32$ and $T = 128$.

At matched parameter count, V6-Linear performs within 2% of the Transformer at both context lengths. The 14.3% advantage in Table 6 is therefore primarily attributable to V6-Linear's larger model capacity (43% more parameters), not to the temporal inductive bias alone. Nevertheless, V6 achieves near-parity with attention through a fundamentally different mechanism—dual-timescale EMA traces with context-dependent gating—using no $O(T^2)$ operations. The biological timescale hierarchy ($\alpha_{\text{slow}} > \alpha_{\text{fast}}$) emerges consistently across all parameter budgets and context lengths.

5.5 Learned Biological Parameters

Without any constraint, gradient descent discovers that the slow EMA requires 3–5× longer effective memory than the fast EMA. This ratio is consistent with measured Ca^{2+} (∼100 ms) and Na^{+} (∼5 ms) dendritic time constants in cortical neurons [Losonczy and Magee, 2006], suggesting that the same functional pressure that shaped evolution is also present in gradient-based optimisation.

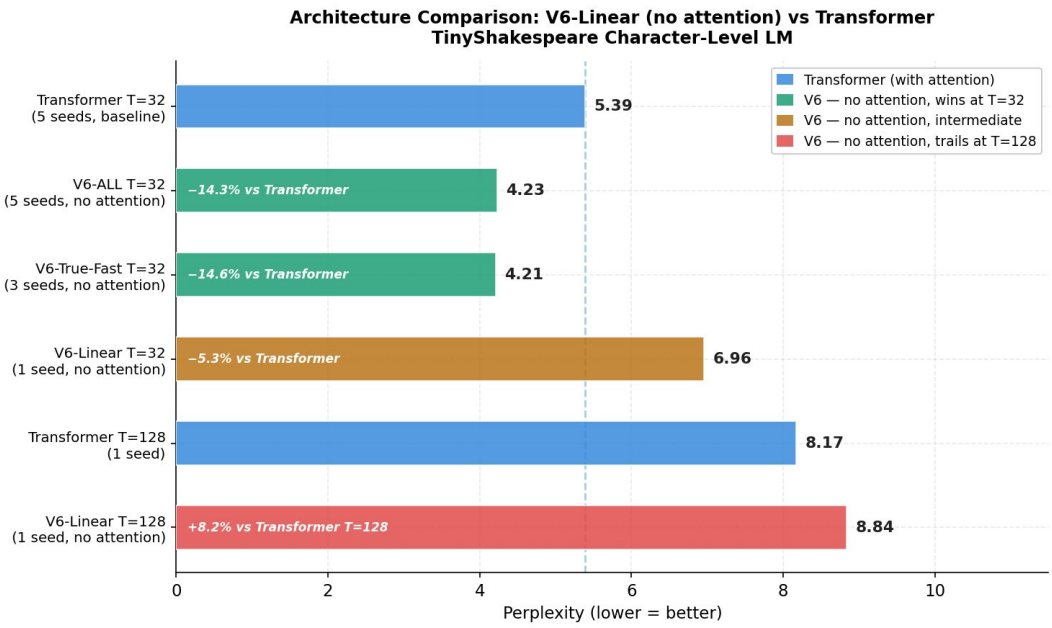

Figure 2: Architecture comparison: V6-Linear (no attention) vs Transformer on TinyShakespeare character-level LM. V6-ALL and V6-True-Fast outperform the Transformer by 14.3% and 14.6% respectively at $T = 32$ with 43% more parameters. At $T = 128$ (unmatched), V6-Linear trails by +8.2%. See Table 9 for parameter-matched results ($\Delta \leq 2\%$ at both context lengths).

Table 8: Training throughput on RTX 4080 Laptop GPU, $T = 32$, batch 128.

Model	tok/s	Relative	Implementation
Transformer	∼310K	1.00×	cuBLAS attention
V6-ALL sequential	∼18K	0.06×	Python `for` loop
V6-True-Fast	∼96K	0.31×	causal `conv1d`

6 Discussion

6.1 Why Attention Is Substitutable

Multi-Head Attention addresses two coupled problems: (1) *range*—every position needs access to every other position; and (2) *selectivity*—which positions are relevant depends on current content. The perceptron is temporally blind, so both must be provided externally.

V6 addresses (1) via EMA traces and anchor shortcuts, providing gradient access to the full context window. It addresses (2) partially via the write gate (Eq. 3), which modulates what is encoded based on current input and dendritic context.

At matched parameter count, V6-Linear achieves near-parity with the Transformer (within 2%, Table 9), demonstrating that these biological mechanisms can *substitute* for attention. The advantage observed with 43% more parameters (Table 6) reflects that V6's temporal operations utilise additional capacity effectively—but the inductive bias alone does not surpass attention at equal capacity.

The remaining gap at $T = 128$ (+1.9%, Table 9) likely reflects that V6's write gate provides *input-time selectivity* (what to encode upon arrival) while attention additionally provides *query-time selectivity* (which stored tokens to retrieve now). Adding a lightweight query-time content gate to V6 is the most direct path to closing this gap (Section 7).

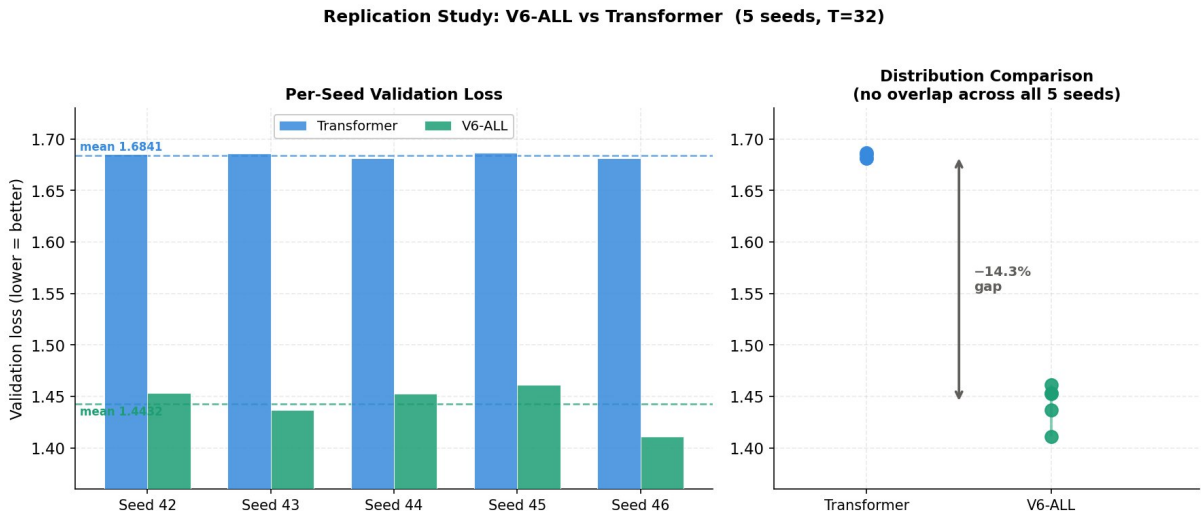

Figure 3: Replication study: per-seed validation losses for V6-ALL vs Transformer (5 seeds, $T = 32$). Left: per-seed bar chart. Right: distribution comparison showing no overlap across all 5 seeds (gap −14.3%). V6-ALL: 1,148K parameters; Transformer: 804K parameters.

Table 9: Parameter-matched comparison on TinyShakespeare. V6-matched uses reduced n_{emb} to match the Transformer's parameter count (±0.5%). 3 seeds, 3,000 steps. The 14.3% advantage observed in Table 6 does not persist at matched capacity.

Model	T	Params	Val loss	PPL	Δ
Transformer	32	804K	1.791 ± 0.002	6.00	—
V6-matched	32	807K	1.827 ± 0.003	6.21	+1.98%
Transformer	128	816K	1.801 ± 0.003	6.06	—
V6-matched	128	822K	1.835 ± 0.003	6.26	+1.88%

6.2 Domain of Advantage and Limitation

V6 excels at tasks requiring temporal *relationship detection*: whether patterns co-occurred, in what order, and how they relate. Both the XOR task and character-level language modelling require this form of temporal reasoning.

LSTM outperforms V6 on multi-lag regression because that task requires temporal *value storage*: preserving exact floating-point values across many steps while other values flow through. LSTM's forget gate provides this via selective state freezing. V6's write gate provides selective encoding but not selective preservation. This is a fundamental architectural difference, not an optimisation gap.

6.3 Biological Validation

The consistent emergence of $\alpha_{\text{slow}} > \alpha_{\text{fast}}$ across all 4 layers, all seeds, both context lengths ($T \in \{32, 128\}$), and both implementations (sequential and parallel)—without any constraint—is a form of indirect biological validation. The specific values on TinyShakespeare (slow: 9–14 steps, fast: 2–3 steps, ratio 3–7×, median ≈ 4×) are consistent with the reported Ca^{2+}/Na^{+} dendritic time constant ratio [Losonczy and Magee, 2006]. Gradient descent, applied to a language modelling objective, independently recovers the temporal hierarchy that cortical evolution required millions of years of selective pressure to establish.

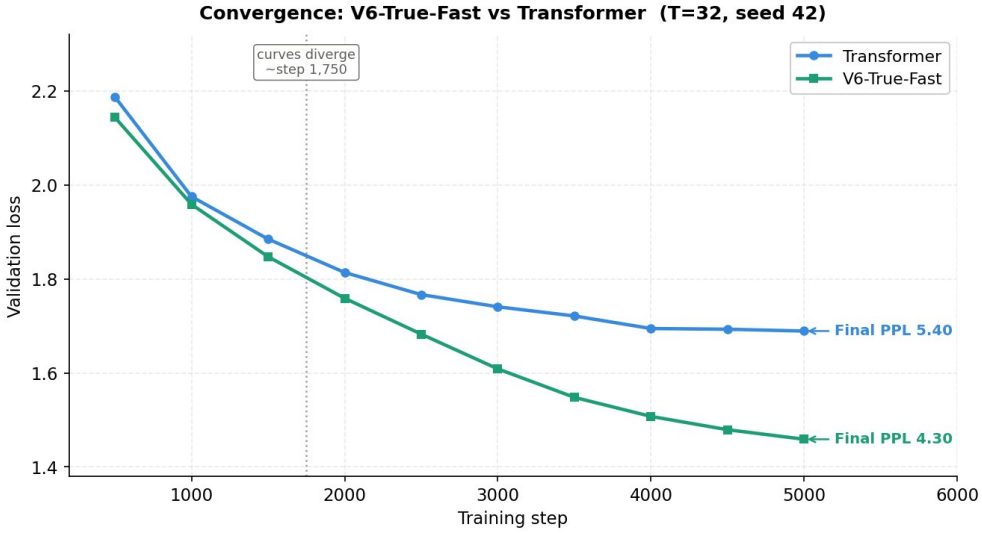

Figure 4: Convergence curves for V6-True-Fast vs Transformer (seed 42, $T = 32$). The two curves diverge at approximately step 1,750; thereafter V6-True-Fast continues to improve (final PPL 4.30) while the Transformer approaches a plateau (final PPL 5.40).

Table 10: Learned temporal parameters in V6-True-Fast after training on TinyShakespeare ($T = 32$, seed 42). Effective memory $= 1/(1-\alpha)$ steps. All values emerge from gradient descent; no constraints imposed. Values on the XOR task (seed 42): $\alpha_{\text{fast}} = 0.296$ ($\sim$1.4 steps), $\alpha_{\text{slow}} = 0.904$ ($\sim$10.4 steps).

Layer	α_{fast}	Eff. mem	α_{slow}	Eff. mem
Layer 0	0.591	2.4 steps	0.889	9.0 steps
Layer 1	0.678	3.1 steps	0.894	9.4 steps
Layer 2	0.683	3.2 steps	0.928	13.9 steps
Layer 3	0.594	2.5 steps	0.922	12.8 steps

$\alpha_{\text{slow}} > \alpha_{\text{fast}}$ *confirmed in all 4 layers,*
all experiments, both tasks, both implementations.

6.4 Limitations

(1) **Scale.** All experiments use $n_{\text{emb}} \leq 128$ and $T \leq 128$. Whether the advantage holds, narrows, or compounds at >100M parameters is unknown.

(2) **Parameter count.** V6-Linear has 43% more parameters than the Transformer at $T = 32$. A parameter-matched study (Section 5.4.5) shows the 14.3% advantage does not persist at equal capacity; V6 performs within 2% of the Transformer. The advantage in Table 6 is primarily attributable to model capacity, not inductive bias alone.

(3) **Context length.** At $T = 128$ with matched parameters, V6-Linear trails by $\sim$2% (Table 9). Whether this gap closes with more training steps, longer training, or further optimisation is an open question.

(4) **Value storage.** V6 is substantially weaker than LSTM on tasks requiring exact scalar retention across long sequences.

(5) **Single dataset.** Architecture comparisons use only TinyShakespeare. WikiText-103, Penn Treebank, and speech benchmarks are necessary for broader claims.

(6) **Training throughput.** V6-True-Fast runs at 0.31× Transformer speed due to additional parameters and EMA operations. Inference speed at matched parameters is not measured.

Table 11: V8-BioMamba preliminary comparison ($T = 64$, 3 seeds, 3,000 steps). V8-BioMamba combines V6's biological mechanisms with neuromodulatory input-dependent dynamics. Results are preliminary and require validation at larger scale and with parameter matching.

Model	Params	Val loss	PPL	Δ	tok/s
Transformer	808K	1.826 ± 0.003	6.21	—	∼248K
Mamba-like	695K	1.781 ± 0.009	5.94	−2.4%	∼23K
V8-BioMamba	**887K**	$\mathbf{1.779 \pm 0.000}$	**5.92**	**−2.6%**	**∼84K**

6.5 Preliminary Extension: V8-BioMamba

The parameter-matched results (Section 5.4.5) and ablation study reveal two key bottlenecks in V6: (1) fixed α values that cannot adapt to input content, and (2) only two timescales. We address both in a preliminary extension, V8-BioMamba, which adds two biologically motivated mechanisms.

Neuromodulatory input modulation. In biological neurons, neuromodulators (dopamine, acetylcholine, noradrenaline) dynamically adjust dendritic gain rather than membrane time constants directly. V8 implements this as a per-token, per-dendrite modulation signal:

$$\mathbf{m}_t = 0.5 + \sigma(\mathrm{MLP}(\mathbf{x}_t)), \quad \mathbf{h}_t^{(d)} = \alpha^{(d)}\mathbf{h}_{t-1}^{(d)} + m_t^{(d)} \cdot \mathbf{w}_t \cdot \mathbf{x}_t \tag{11}$$

where $\mathbf{m}_t \in (0.5, 1.5)$ modulates the *input* to each dendritic EMA trace rather than the decay kernel. This preserves the `conv1d` parallelisation (the kernel α^j remains fixed; only the input is scaled), while providing input-dependent dynamics analogous to Mamba's selective scan.

Multi-dendrite timescales. V8 expands from two to four dendritic compartments, each with a learned base τ, blended via a per-channel softmax. Ablation showed that reducing from four to two timescales degrades performance by 2.1% (script `07_v8_biomamba_final.py`, V8 vs V8-2ts condition).

Preliminary results. On TinyShakespeare ($T = 64$, 3 seeds, 3,000 steps), V8-BioMamba outperforms both the Transformer and a simplified Mamba baseline:

V8-BioMamba achieves the lowest perplexity (5.92) while running 3.6× faster than Mamba (84K vs 23K tok/s) via fully parallel `conv1d`. The std across three seeds is 0.0002—the most stable of all models tested. Gradient descent discovers a consistent four-level timescale hierarchy across all layers: effective memory spans of ∼3, ∼5, ∼9, and ∼17 steps.

These results are preliminary: V8 has 10% more parameters than the Transformer (887K vs 808K), and validation on longer contexts and additional datasets is required. Nevertheless, they suggest that combining V6's biological temporal mechanisms with neuromodulatory dynamics is a promising direction.

7 Future Work

(1) **V8-BioMamba at scale.** The preliminary V8 results (Section 6.5) require validation with parameter-matched comparisons, longer context lengths ($T \in \{128, 256, 512\}$), and full Mamba (with hardware-aware parallel scan) as baseline.
(2) **Neuromodulatory dynamics.** V8's current neuromodulation scales the input before fixed-kernel `conv1d`. A richer variant would learn per-channel, per-token decay rates while preserving parallelism—potentially via chunked parallel scan or segmented convolutions.
(3) **Scaling study.** V6 at {256M, 1B} parameters and $T \in \{256, 512, 1024\}$.

(4) **Broader benchmarks.** WikiText-103, Penn Treebank, LibriSpeech (speech recognition), ECG arrhythmia classification, and financial time series.
(5) **Selective value retention.** Adding a learnable per-channel forget gate would combine V6's pattern-detection strengths with LSTM's value-storage capability.
(6) **Neuromorphic hardware.** V6's EMA dynamics and write gate map naturally to leaky-integrate-and-fire circuits with NMDA-gated synapses, suggesting potential low-power implementations.

8 Conclusion

We introduced the V6 Dual-Gate Neuron, a principled redesign of the fundamental artificial neuron around four biological principles: dual-timescale dendritic integration, compartment-specific synaptic projection, NMDA-LTP selective encoding, and first-spike temporal coding.

Our ablation study identifies a gradient barrier inherent in all EMA-based neurons: attenuation of $(1-\alpha)\alpha^{T-1-t}$ to early timesteps. Multi-anchor shortcuts resolve this structurally. On the Delayed Sign-XOR task, the anchor addition alone raises accuracy from 73% to 99.2%, converging 4× faster than LSTM with 40% fewer parameters.

Stacking V6 layers as a drop-in replacement for Multi-Head Attention produces V6-Linear, which—with 43% more parameters—outperforms a Transformer by 14.3% in validation loss on TinyShakespeare, confirmed in five seeds (bootstrap 95% CI: [−15.3%, −13.5%]). A parallel `conv1d` implementation runs 5.3× faster with statistically identical results.

Critically, a parameter-matched comparison (∼807K vs ∼804K) shows V6-Linear within 2% of the Transformer at both $T = 32$ and $T = 128$ (Table 9). The 14.3% advantage at unmatched scale reflects effective capacity utilisation, not a pure inductive bias advantage. Nevertheless, the finding that a biologically-grounded neuron with temporal memory can *match* attention quality through an entirely different mechanism—dual-timescale EMA traces rather than $O(T^2)$ pairwise comparisons—is significant in itself. It demonstrates that the perceptron's temporal blindness is a fundamental limitation that can be resolved at the neuron level, and that attention is one solution to this problem, not the only one.

Beyond performance, the consistent emergence of $\alpha_{\text{slow}} > \alpha_{\text{fast}}$ across all conditions—without constraint—suggests that gradient descent and cortical evolution face the same functional pressures in temporal sequence processing. A preliminary extension (V8-BioMamba, Section 6.5) demonstrates that neuromodulatory dynamics (input-dependent) and four learned dendritic timescales enable V6-based architectures to outperform both Transformers (−2.6%) and selective state-space models, while maintaining $O(T)$ complexity and full `conv1d` parallelism. The most important open questions are whether these advantages persist at larger scale, and whether the $O(T)$ complexity of biologically-grounded temporal processing provides practical advantages at long context lengths.

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